

PALETTE-Aug: A Simple Texture-Preserving Augmentation for Medical Imaging Domain Generalization

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Abstract

Introduction. Deep learning for medical image segmentation generalizes poorly across imaging modalities, and annotating data for every new domain is prohibitively costly. Synthesis-based augmentation like SynthSeg address this by generating training images from label maps alone, but require dense label maps and discard the source image’s texture. We introduce PALETTE-Aug, a texture-preserving alternative, and ask when texture matters for generalization. **Methods.** We benchmark PALETTE-Aug against five methods on eight segmentation tasks: five whose targets are bounded by an anatomical interface (healthy brain, multi-organ, healthy spine, mandible, pelvis) and three defined by tissue appearance (brain tumor, multiple sclerosis lesions, breast lesions). Each method trains on one modality and is evaluated in-domain and out-of-domain. An ablation isolates the effect of texture preservation. **Results.** PALETTE-Aug achieves the highest Dice on six of the eight tasks and significantly outperforms every competing method overall. In the ablation (six tasks), isolating texture preservation yields +5.3 and +4.2 Dice on two interface-free targets (brain tumor, multiple sclerosis), versus at most +1.4 on interface-bounded ones (healthy brain: -1.9); breast lesions, also interface-free, show no Dice gain. **Conclusion.** Large gains from texture preservation arise only where the target lacks an anatomical interface, and PALETTE-Aug generalizes best overall across eight segmentation tasks.

1. Introduction

Medical image segmentation models are trained on one acquisition setting and are frequently applied to another: a different scanner, sequence, or imaging modality than the one seen during training. MRI contrast is a protocol choice rather than a physical constant, so the same anatomy yields incompatible intensity distributions across sequences, and a network that has only seen T1-weighted brain images can fail outright on T2-weighted or FLAIR images of the same

anatomy [3, 34]. Acquiring and annotating training data for every contrast a model might encounter is not practical: manual segmentation is expensive, and clinical sites rarely share raw data across institutions [33, 51]. The single-source domain generalization setting — train on one labeled dataset in one contrast, generalize to unseen contrasts and modalities without any target-domain data — is therefore the setting that matters for deployment, and the one a growing body of work addresses [36, 45, 55, 57].

The dominant single-source strategy is label-generative domain randomization, popularized by SynthSeg [3]. Given a dense anatomical label map — a voxel-wise parcellation of the whole image, not the sparse structure-of-interest annotation a segmentation dataset normally provides — intensities are resampled from a per-label Gaussian mixture, producing images of arbitrary, unrealistic contrast that force the network to ignore contrast and rely on shape and layout [3]. This is effective, but carries two structural costs. First, it is *label-driven*: the dense parcellation it resamples from is a richer annotation than the task itself requires, so generating synthetic contrast costs exactly the expensive labeling the method aims to reduce reliance on. Second, and less discussed, it *does not preserve texture*: replacing every labeled region with piecewise Gaussian noise discards the real gradients, partial-volume boundaries, and unlabelled fine structure the acquired scan contains. Layout survives; texture does not.

We ask whether discarding texture is necessary for contrast robustness, or only convenient once a dense label map is assumed. Our central finding is that the answer depends on the *task*: large gains from preserving real intensity structure occur only where the target is defined by tissue appearance, not where an anatomical interface separates it from its surroundings. Prior work shows that texture carries diagnostic signal in such tasks: texture features separate MS lesions from normal-appearing white matter [56], and diffuse glioma infiltrates beyond every margin visible on conventional MRI [17, 47]. We show that this distinction determines where a texture-preserving augmentation helps substantially.

We exploit it with **PALETTE-Aug**, a single-source augmentation that needs no dense label map. **PALETTE-Aug** groups the foreground intensities of a scan into classes with 1-D k -means, splits each class into spatial sub-regions with a Voronoi partition, and applies a signed random affine remap to the *real* intensities of each sub-region. Because existing intensities are remapped rather than replaced by noise, edges, gradients and partial-volume detail survive inside each sub-region. Only the contrast between regions is randomized, including sign inversions that no acquisition protocol produces. An optional last step uses labels, and only the sparse task labels that any segmentation dataset already has.

We test the mechanism by intervention. Keeping the partition and the region target means, we replace only the real-intensity remap with SynthSeg-style noise fill, in fourteen training settings. Every large effect of that swap is on a target without a tissue interface (+13.3 and +7.2 Dice on brain tumor, +7.7 on MS lesions, the last for all eight test patients), and no interface-bounded target gains more than 1.4. The converse does not hold: several appearance-defined settings, breast lesions among them, show no effect, so we state the result in one direction only (§4.3). A segmentation-free gradient measurement confirms that the swap changes texture as intended: the noise-fill rungs keep only the partition edges, and the real-fill swap restores the source texture.

We train one nnU-Net [22] configuration, varying only the epoch budget per dataset, and evaluate cross-contrast, cross-modality and cross-dataset transfer on eight tasks: healthy brain, MS lesion, brain tumor, abdominal organ, mandible, breast lesion, spine and pelvis. We compare against a no-augmentation baseline, two variants of SynthSeg (the released dense-label variant and our re-implementation of the unreleased EM variant), SRCSM [48], and Auglab [34], a published cross-modality augmentation pipeline. **PALETTE-Aug** is the best method overall: it has the best Dice on six of the eight tasks and is significantly better than every competitor across tasks, on both Dice and boundary distance. The best competitor switches between Auglab and SRCSM from task to task, while **PALETTE-Aug** stays at or near the top throughout.

In summary, we make the following contributions:

- **A task-level account of when texture preservation matters.** Large gains from preserving real intensity structure occur only where the segmentation target has no anatomical interface separating it from its surroundings. We test this with a controlled intervention rather than asserting it post hoc, and report where it does not extend.
- **A causal ablation isolating texture.** Swapping only the real-intensity remap for noise fill, with the partition fixed, confines every large effect to no-interface targets in fourteen training settings. A segmentation-free gradient mea-

surement confirms that the swap changes texture as intended.

- **PALETTE-Aug**, the single-source contrast augmentation that exploits this, requiring no dense anatomical label map and using only the sparse task labels a segmentation dataset already provides.
- **Breadth evidence.** Across cross-contrast, cross-modality and cross-dataset transfer on eight tasks and sixteen datasets, with one configuration reused throughout (Spine was run on a separate pipeline), **PALETTE-Aug** is best on six tasks and significantly better than every competitor across tasks. Where it does not win we report whether the method ahead is actually ahead: on healthy brain it is a tie, on mandible a real loss.

2. Related Work

2.1. Domain generalization for medical segmentation

Domain generalization (DG) targets unseen distributions without access to target data, unlike unsupervised domain adaptation, which assumes unlabelled target images and solves an easier problem [44]. Most medical DG methods are *multi-source*, interpolating across several labeled domains by meta-learning, feature alignment, frequency-space exchange, multi-frequency architectures, foundation-model scale, or diffusion-based stylization [2, 30, 33, 35, 42, 51] – all of which still rely on source diversity. We instead work in the *single-source* regime — one labeled dataset, no target data, no second source – the setting that matters when target-domain data is unavailable at training time, which is routine in clinical deployment [36, 57]. This rules out cross-domain interpolation and makes architecture-coupled methods orthogonal to our contribution: our mechanism acts at the input level and leaves the segmentation network a fixed nnU-Net [22].

2.2. Synthesis-based contrast-agnostic training

The dominant single-source strategy is label-generative domain randomization, introduced by SynthSeg [3]: intensities are sampled from a per-label Gaussian mixture over an anatomical segmentation, yielding images of arbitrary contrast that force contrast-invariance. This family carries two structural costs: it is *label-driven* (a dense anatomical label map is required per subject – SIAM shifts the dependence from label *quantity* to *quality* with curated templates, but each still needs a full dense parcellation [49]), and it is *texture-destroying*, since per-label Gaussian sampling replaces real gradients, partial-volume detail and unlabelled structure with piecewise noise, retaining layout but not texture. The lineage has been specialized to white-matter and MS lesions and to atlas-based expectation-maximization [6, 7, 28]. Among SynthSeg’s own variants

we take the original (*noEM*), which resamples intensities from a dense anatomical label map and never looks at the acquired scan, and the *EM* variant, which instead fits its per-class intensity model to the real image by expectation-maximization and so needs only the sparse task labels a segmentation dataset already provides, not a dense parcellation. The EM variant has no released implementation; ours follows the mechanism described in the original paper [3].

2.3. Texture as a segmentation signal

That local texture, not only intensity level, carries information a segmentation model can use is established independently of us: second-order gray-level statistics discriminate image content beyond simple intensity [18], and quantitative texture features from tumor regions carry prognostic signal across CT, MRI and PET [27]. Closer to our setting, Kang et al. disentangle structure from texture in cross-domain image translation with a texture co-occurrence loss, targeting the same failure mode we do — naive translation distorts fine texture and degrades downstream segmentation [24]; a texture-fidelity loss has likewise been added to GAN-based lesion augmentation [15]. That some targets resist boundary-based methods for lack of a sharp edge, and are better served by texture-based segmentation, has been argued directly for diffuse white-matter lesions [26]. None of this prior work, to our knowledge, connects texture preservation in an *augmentation* to *which* segmentation targets it helps; that connection, tested causally, is our contribution.

2.4. Single-source intensity and appearance augmentation

Where the first line of single-source methods synthesizes from a label map, a second manufactures appearance diversity directly from the real image, with no generative label model. General-purpose libraries stack intensity, spatial and artifact transforms on an nnU-Net trainer; the recent Auglab pipeline [22, 34] is one such GPU-accelerated framework and is the strongest competitor across our tasks, and BigAug stacks a comparable set of coarse intensity, quality and spatial perturbations under the same single-source DG protocol we use [57]. Our method inserts the PALETTE-Aug transform as one additional operator inside the Auglab pipeline, so the comparison against Auglab directly isolates PALETTE-Aug’s contribution. Closer to our operator are random-filter methods, which vary appearance with an untrained nonlinear filter: GIN applies randomly re-initialized convolutions for label-free appearance variants [36]; related variants establish the random-filter principle for natural images, perturb intensities per class using the task mask, add learned diffusion generators, or perturb the Fourier amplitude spectrum instead of the image itself [8, 31, 39, 45, 55, 58]. Our operator instead parti-

tions the foreground into intensity classes and Voronoi sub-regions and applies a signed *affine* remap of the real intensities within each, invertible and gradient-preserving by construction. The closest peer is SRCSM [48] (semantic-aware random convolution with source matching), which also localizes its perturbation region-wise in true 3-D and, like PALETTE-Aug, layers exactly one label-consuming component over an otherwise label-free step: at training time it draws a fresh untrained convolution per anatomical class from the ground-truth map (needing a Gaussian smoothing pass to control the boundary-contrast artifacts this introduces), and at test time applies a label-free histogram matching onto the source distribution. The difference from PALETTE-Aug is not label use versus none, but *how* that step transforms the signal: an untrained nonlinear convolution versus a deterministic affine remap that leaves relative intensity structure inside each region intact.

2.5. Cross-contrast evaluation protocol

We evaluate a model trained on a single real input modality on every held-out modality of the *same* segmentation task, including the training modality itself, across eight tasks spanning brain, abdominal, dental, breast, spine and pelvic anatomy in MRI, CT and CBCT. This single-sequence-in, every-sequence-out design is not unprecedented — Auglab’s own evaluation follows the identical protocol for spine segmentation [34] — and our contribution is breadth (eight tasks, not one) and a further axis: wherever an external cohort exists, it is pooled into its task as additional held-out contrasts from other sites, so the training dataset’s own test set is one point among several rather than the only held-out evidence. For mandible, which is acquired as CBCT only, that cross-dataset transfer is the entire held-out axis.

3. Method

We describe PALETTE-Aug (§3.1), how it is applied during training (§3.2), and the causal ablation used to isolate texture preservation as the source of its effect (§3.3).

3.1. The PALETTE-Aug transform

Let x denote the intensity volume of a training scan, min-max normalized to $[0, 1]$, and $F \subset \Omega$ its foreground voxels (an intensity threshold; $x_v > 0.01$ in our implementation). PALETTE-Aug applies four steps at every training iteration, illustrated end-to-end in Fig. 1 on one real CHAOS (abdominal MRI) and one real ON-Harmony (brain MRI) volume. Panels (b)–(d) show each step’s flat target values; only (e), the training input, shows the texture the transform produces.

1. Intensity partition. We run 1-D k -means on the foreground intensities $\{x_v : v \in F\}$ with k sampled per

iteration from $\{2, \dots, 6\}$, producing k intensity classes $C_1, \dots, C_k$ (Fig. 1, panel (b)). This partition is unsupervised and requires no anatomical label map: it groups voxels by intensity alone, so the same operator applies unmodified to any input image regardless of its modality or contrast — CT, T1, T2, FLAIR, or otherwise. This step is occasionally skipped in favor of a single global remap (step 3 below, with $C = 1$); per-step probabilities are set once and never tuned per dataset, and are listed in Sec. 7.

2. Spatial sub-parcellation. Each intensity class C_i is further subdivided into spatial sub-regions by a Voronoi partition seeded with points sampled within C_i (Fig. 1, panel (c) shows the combined effect of this step and the remap of step 3), giving sub-regions $R_{i,1}, \dots, R_{i,m_i}$ that are contiguous in space as well as similar in intensity. This second partition breaks the spatial correlation of a purely intensity-based grouping: two voxels with the same intensity in different parts of the anatomy can land in different sub-regions and receive independent contrast perturbations. Each class is usually, but not always, sub-split this way (Sec. 7).

3. Signed affine remap. For each sub-region R , let $\bar{x}_R$ denote its mean intensity, $\mu_R \sim U(0, 1)$ a resampled target mean, and draw $\alpha_R \sim \pm U(0.5, 2.0)$ (a magnitude in $[0.5, 2.0]$ with an independently sampled sign). Every voxel $v \in R$ is remapped as

$$y_v = \mu_R + \alpha_R (x_v - \bar{x}_R). \quad (1)$$

Because Eq. (1) is a per-region affine function of the *real* intensity x_v , it preserves the relative structure of the signal within each region: edges, gradients, and partial-volume transitions between adjacent voxels are scaled, not replaced. Only the region-to-region contrast relationship is randomized, and $\alpha_R < 0$ additionally allows local contrast inversions that no acquisition protocol produces on its own, widening the space of contrasts the network sees beyond what any single real sequence could supply. Background voxels are zeroed at this step (and remain zero through the rest of the pipeline). The result is occasionally Gaussian-blurred, mostly not (Sec. 7).

4. Anatomical-label affine remap (optional). Steps 1–3 are unsupervised, so the k -means/Voronoi partition need not respect a true anatomical boundary: when two adjacent tissues share similar intensity, step 3 can by chance assign them the same target contrast and erase that boundary rather than randomizing it. This step corrects for that: for each real anatomical label ℓ present, we resample a fresh (μ_ℓ, α_ℓ) and apply Eq. (1) *again*, restricted to ℓ 's voxels, **on top of the step-3 output** rather than the original voxels (Fig. 1, panel (d)) – so it composes with, rather than replaces, the unsupervised remap, and ℓ 's voxels still inherit the internal structure

steps 1–3 imposed on them. Because label assignment is independent of the unsupervised partition, this reintroduces a full-strength contrast discontinuity at the true label border regardless of whether step 3 preserved it, sharpening the boundary cue available to the network – not the underlying anatomy. It consumes only the labels the segmentation task already requires, so it needs no additional annotation, and is optional: disabling it (`label_remap_prob=0`) degrades PALETTE-Aug gracefully to steps 1–3 alone, e.g. on an unlabelled dataset. A final foreground z-score normalization follows an optional second blur pass, giving the training input shown in panel (e) of Fig. 1. Per-step probabilities are set once, never tuned per dataset, and listed in full in Sec. 7.

3.2. Training protocol

PALETTE-Aug is applied on-the-fly as one contrast-augmentation operator inside an existing GPU-accelerated nnU-Net augmentation pipeline [22, 34], alongside that pipeline's standard spatial and artifact augmentations; it adds no learned parameters and no measurable training-time overhead. A single configuration – the same k -range, sub-region count, blur schedule, and α range (Sec. 7) – is tuned once and reused unchanged across every dataset and modality in Sec. 4; no dataset receives its own hyperparameter sweep.

3.3. Causal ablation: isolating texture preservation

PALETTE-Aug differs from label-generative synthesis (e.g. SynthSeg) along two axes at once: the partition it perturbs (unsupervised k -means + Voronoi vs. a supplied anatomical label map as the sole generative substrate) and the perturbation itself (an affine remap of real intensities vs. re-sampling from a per-label Gaussian mixture). Comparing PALETTE-Aug to SynthSeg directly therefore cannot attribute an advantage to either axis in isolation.

We construct a controlled variant, *noise-fill*, with the same k -means/Voronoi partition and the same random target mean μ_R for each sub-region as Eq. (1). Instead of remapping the real intensities, it fills each sub-region with SynthSeg-style Gaussian noise, $\mu_R + \sigma_R \epsilon$ with $\epsilon \sim \mathcal{N}(0, 1)$ and σ_R drawn uniformly from $[0.05, 0.25]$ per sub-region. The partition and the target means are PALETTE-Aug's own, so a difference between the two isolates whether the real texture inside each region is kept or discarded. We report this comparison separately for targets with and without a tissue interface, assigned by anatomy (Sec. 4); the texture hypothesis predicts a larger gap on the latter.

4. Experiments

4.1. Setup

Protocol. We train a single real-intensity source modality per dataset and evaluate on held-out modalities of the

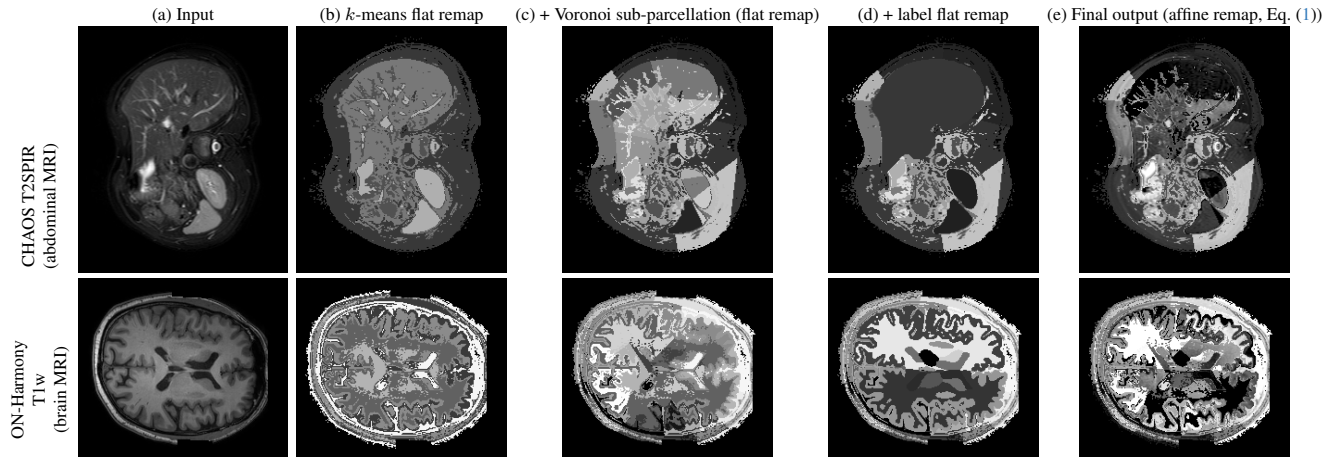


Figure 1. The PALETTE-Aug pipeline on one real CHAOS T2SPIR volume (top) and one real ON-Harmony T1w volume (bottom); every panel is a deterministic function of the same real volume (no external model), min-max normalized to $[0, 1]$ for display only. Panels (b)–(d) paint each region, sub-region, or label with its precomputed target mean μ from Eq. (1), isolating that flat value from the texture applied on top of it; panel (e) is the actual training input, produced by applying the full signed affine remap of Eq. (1) with real texture (not a flat fill) at both the region and, where available, anatomical-label level.

same segmentation task, using one fixed nnU-Net [22] configuration across all settings. No target-domain data, of any modality, is seen during training. Our full method, denoted **Ours** throughout, is PALETTE-Aug applied as one additional contrast-augmentation operator inside the Auglab pipeline [34]; the *Auglab* baseline is the same pipeline with the PALETTE-Aug operator removed, so the two differ by exactly the PALETTE-Aug transform. Every method is trained with the identical nnU-Net schedule — SGD with Nesterov momentum (0.99), the nnU-Net `polyLR` decay from an initial rate of 10^{-2} , 250 mini-batches per epoch, and the combined Dice-plus-cross-entropy loss — and differs only in the augmentation applied at the input. The epoch budget is fixed per dataset and shared by every method on that dataset (2500 for Brats-GLI, 2000 for ON-Harmony and Open-MS, 1000 for ToothFairy2 and I-SPY2, 200 for CHAOS and Pelvis; the Spine models were trained on a separate pipeline with the same method suite, Sec. 8), and each method is trained as three models on three train/validation folds and all three are scored on the same held-out test set. All significance tests are paired, use the patient as the unit, and are Holm-corrected [20]; full test construction, configuration details, and per-dataset descriptions (subject counts, acquisition protocols, and label provenance) are given in Secs. 7 to 9.

Datasets. We evaluate on eight tasks drawn from sixteen datasets (Tab. 1): abdominal organs, healthy brain, brain tumor, MS lesions, mandible, breast lesions, spine and pelvic musculoskeletal anatomy, across MRI, CT and dental CBCT. Brain tumor and MS lesions use Brats2024-GLI [10] and Open-MS [29]. Every task trains on a single source

modality at a time and is tested on every other contrast of that task *and* the training contrast itself — eighteen source-modality settings in total; no target modality is seen during training. Wherever an external cohort exists, its contrasts are pooled into that task as further held-out test columns rather than reported as a task of their own: CHAOS [25] transfers to CT from AMOS [23] and SLIVER07 [19]; Mandible (ToothFairy2 [5], CBCT only) to the CT and MR of HaN-Seg [38] and the CT of PDDCA [40], which makes it the one task whose held-out axis is entirely cross-dataset; Breast (I-SPY2 [32]) to Duke-Breast-MRI [13, 41], I-SPY1 [21] and ACRIN-6698 [37], the last contributing diffusion-weighted MRI, a contrast neither training modality resembles; and Spine (a CT/Dixon cohort) to the T1w and T2w of SPIDER [50]. Pelvis is built from the TotalSegmentator CT and MRI releases [9, 53], two unpaired cohorts. ON-Harmony [52] reference labels are silver-standard, obtained on T1w with FreeSurfer [12] and registered to the other contrasts.

Baselines. SynthSeg comes in two variants, which we refer to throughout by how they obtain their per-class intensity model: *noEM*, the released variant, which reads it off a dense anatomical label map; and *EM*, which estimates it from the image by expectation-maximization and therefore needs no dense parcellation. We compare against five methods. (1) *No-augmentation*, the nnU-Net default training recipe. (2) *SynthSeg-noEM* [3], the original label-generative variant (the implementation released by the authors), which synthesises each training image from a dense anatomical label map alone, ignoring the acquired scan entirely. (3) *SynthSeg-EM*, the variant described in the original work

Task	Dataset	Train modalities	Test modalities
Multi-Organ	CHAOS	T1-in, T2-SPiR	T1-in, T1-out, T2-SPiR, CT
	AMOS	–	CT, MRI
	SLIVER07	–	CT
Brain Glioma	Brats2024-GLI	T1n, T1c, T2w, FLAIR	T1n, T1c, T2w, FLAIR
Brain MS	Open-MS	FLAIR, T1w	FLAIR, T1w, T2w
Healthy Brain	ON-Harmony	T1w, T2w, DWI	T1w, T2w, BOLD, DWI, EPI, GRE
Mandible	ToothFairy2	CBCT	CBCT
	HaN-Seg	–	CT, MR-T1
	PDDCA	–	CT
Breast	I-SPY2	T1-CE, T2w	T1-CE, T2w
	Duke-Breast-MRI	–	T1-CE, pre-contrast
	I-SPY1	–	T1-CE, pre-contrast
	ACRIN-6698	–	DWI
Spine [§]	CT/Dixon cohort	CT, Dixon in	CT, Dixon fat/in/water
	SPIDER	–	T1w, T2w
Pelvis	TotalSegmentator	CT, MRI	CT, MRI

Table 1. Per-task dataset roster. Rows marked “–” are cross-dataset transfer targets only, never trained on; their contrasts are extra held-out test columns of the task they sit under, not tasks of their own. Mandible’s held-out axis is entirely cross-dataset, since ToothFairy2 acquires CBCT alone. Pelvis’s CT and MRI come from different patients. [§]Spine results were computed on a separate pipeline with the same method suite except SRCSM (Sec. 8). Subject counts, acquisition protocols, and label provenance are in Sec. 8.

but never released publicly, which we re-implement; unlike SynthSeg-noEM it does not require a dense anatomical label map, but it is not dense-label-free either — it is compatible with and relies on the sparse task labels naturally available in a segmentation dataset, which it uses to estimate its per-class intensity model. (4) *SRCSM*, a semantic random-convolution contrast-synthesis augmentation, included as a second strong synthesis-based competitor that is neither label-generative nor a member of the Auglab family. (5) *Auglab*, the published GPU-accelerated augmentation pipeline of Molinier et al. [34]. Auglab is a prior method in its own right, not an ablation of ours, and we report it among the competitors accordingly; our full method (Sec. 4.1) applies PALETTE-Aug *on top of* the Auglab pipeline, so the PALETTE-vs-Auglab comparison also functions as the add-one-component test isolating the contribution of the PALETTE-Aug transform.

4.2. Main comparison

PALETTE-Aug is the best method overall. It has the best Dice on six of the eight tasks. Across tasks it is significantly better than every competitor ($p < 10^{-10}$ each, Tab. 2). Within single tasks it significantly outperforms every competitor on four tasks (Brats-GLI, Open-MS, Breast, Spine), three or four of five on CHAOS, Mandible and Pelvis, and two of five on ON-Harmony. The cross-task test treats our eight tasks as fixed. Treating each task as one unit instead, which is the question of how a method fares on a new task, PALETTE-Aug beats Auglab on seven of eight tasks (one-sided sign test $p = 0.035$) and SRCSM on five

of the seven tasks where SRCSM was run ($p = 0.23$).

SynthSeg-noEM samples intensities from a dense anatomical label map. It works only where such maps exist and falls below the no-augmentation baseline on six of the eight tasks. SynthSeg-EM, which estimates its intensity model from the task labels, closes most of that gap. SRCSM also falls below the baseline on Breast.

No competitor is reliably second. The best competitor alternates between Auglab (Brats-GLI, Open-MS, Breast, Spine) and SRCSM (CHAOS, ON-Harmony, Mandible, Pelvis), and SRCSM fails on Breast (16.4 Dice). Picking the right one would require knowing the answer in advance. PALETTE-Aug is first or second on every task but one.

Where PALETTE-Aug does not win. The one-sided test above cannot show that a competitor is ahead, so on these tasks we also ran the reverse test on the same patient-level design (Holm-corrected across competitors within the task). On ON-Harmony SRCSM leads by 0.1 Dice and neither method is significantly ahead. On Mandible SRCSM (+1.3 Dice, $p = 6 \times 10^{-4}$) and Auglab (+0.6, $p = 2.5 \times 10^{-3}$) are significantly ahead, and PALETTE-Aug is third. In the Mandible ablation the texture step is significantly positive, and the drop coincides with adding the Auglab pipeline on top (Tab. 10).

Boundary distance. HD95 is reported in the supplement (Tab. 8). PALETTE-Aug has the lowest HD95 on five of the eight tasks (Brats-GLI, ON-Harmony, Open-MS, Spine, Pelvis) and is significantly better than every competitor across tasks. It ties Auglab on Breast, trails SRCSM on CHAOS without a significant gap, and is significantly behind SRCSM, Auglab and the baseline on Mandible.

4.3. Causal ablation: when does texture matter?

PALETTE-Aug’s partition could help on its own, without the texture-preserving remap. We test this with a ladder in which each rung adds one ingredient: baseline $\rightarrow +k$ -means $\rightarrow +$ label remap $\rightarrow +$ Voronoi $\rightarrow$ real-intensity fill $\rightarrow +$ Auglab. The three partition rungs fill every region with label-style Gaussian noise, as SynthSeg does. All rungs run the same pipeline (Sec. 3) in the same order and only switch steps on. The step from +Voronoi to real fill changes only the fill: the partition, the number of sub-regions, the distribution of region target means and both blur passes are the same on each side. A change at that step is therefore due to preserving the real texture.

Fig. 2 and Tab. 3 support a one-directional result. Every large gain from real texture is on a target defined by tissue appearance rather than by an anatomical interface: Brats-GLI T1c (+13.3 Dice) and T1n (+7.2), both significant after correction over the fourteen rows, and Open-MS

	Brats-GLI	CHAOS	ON-Harm.	Open-MS	Mandible	Breast	Spine	Pelvis	p vs ours
Baseline	23.4	29.3	35.0	21.0	59.2	35.0	48.5	29.8	$< 10^{-10}$
SynthSeg-noEM	8.8	59.6	64.8	4.3	53.6	16.5	47.5	15.0	$< 10^{-10}$
SynthSeg-EM	43.1	83.0	66.3	33.9	76.3	39.6	80.6	62.5	$< 10^{-10}$
SRCSM	41.5	83.9	66.6	33.0	79.0	16.4	—	65.3	$< 10^{-10}$
Auglab	44.8	83.3	66.1	34.3	78.3	40.1	80.9	64.3	$< 10^{-10}$
PALETTE-Aug (ours)	45.9	85.0	66.5	38.5	77.7	41.2	82.3	65.6	—
competitors beaten ($p < 0.05$)	5/5	4/5	2/5	5/5	3/5	5/5	4/4	3/5	

Table 2. Task-level Dice (%). Each column pools that task’s held-out contrasts, all of its source modalities and, where one exists, its external cohorts, pooled by contrast group so that a modality tested by several external sources does not outvote one tested by fewer (Sec. 9). p : Holm-corrected one-sided sign-flip test on the macro-averaged per-patient difference, stratified by task with tasks weighted equally — a test *across* tasks, not an average *of* them; we deliberately print no cross-task mean Dice, since averaging a whole-liver Dice with a breast-lesion Dice yields a number no task’s clinician would recognize. The bottom row counts, per task, the competitors PALETTE-Aug beats with the same test run within that task alone. Spine has no SRCSM run (Tab. 1). HD95 is in Tab. 8.

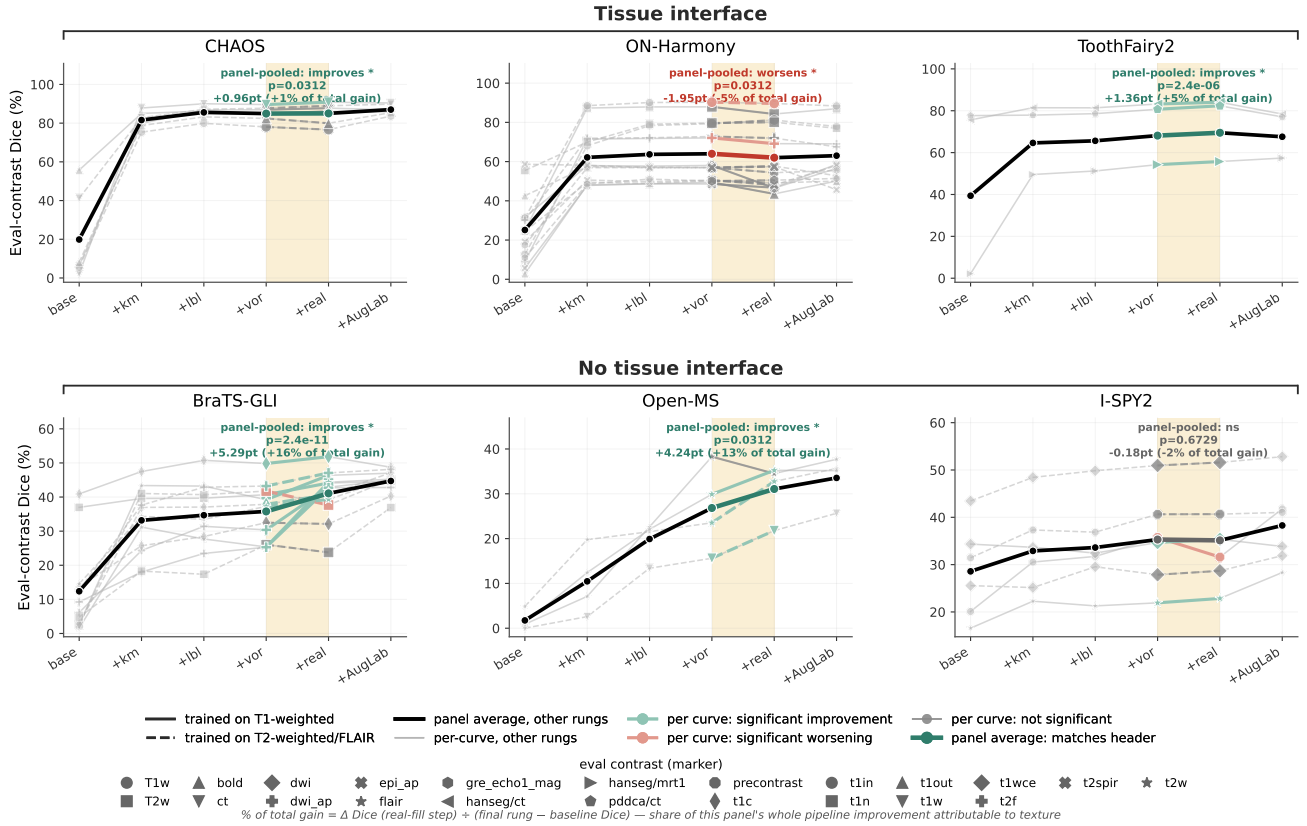


Figure 2. The one-variable test, one panel per task with a ladder, grouped by boundary type. Each thin line is one (training modality, held-out contrast) pair, external cohorts included, and only its real-fill segment (shaded band) is colored, by that pair’s own significance at this one step – teal = significant improvement, red = significant worsening, grey = not significant. The bold black line is the panel average; its real-fill segment is colored to match the panel-level test annotated above it, which pools all of the panel’s training modalities into one paired Wilcoxon test on *patients* (each counted once across contrasts and models), Holm-corrected across the six panels. Two of the three no-tissue-interface panels improve significantly (Brats-GLI $p = 2.4 \times 10^{-11}$, Open-MS $p = 0.031$) and Breast does not move. Among the tissue-interface panels, CHAOS and ToothFairy2 show small significant gains and ON-Harmony a worsening (significant with its eight sessions as units; these come from four subjects). Solid/dashed = trained on a T1-weighted vs. other contrast; marker = held-out contrast. Per-row numbers in Tab. 3; HD95 in Fig. 4 and Tab. 9.

Task	boundary type	Δ Dice	p
CHAOS T1in	interface	+1.07	0.0014
CHAOS T2spir	interface	-0.88	0.17
ON-Harmony T1w	interface [†]	-5.18	0.078*
ON-Harmony T2w	interface [†]	-1.55	0.078*
ON-Harmony DWI	interface [†]	+0.78	0.11*
ToothFairy2 CBCT	interface [‡]	+1.36	5.8e-6
Open-MS FLAIR	no interface	+7.69	0.078*
Open-MS T1w	no interface	+0.79	1.0*
Brats-GLI T1n	no interface	+7.22	3.3e-11
Brats-GLI T1c	no interface	+13.27	5.0e-12
Brats-GLI T2w	no interface	+0.87	0.38
Brats-GLI FLAIR	no interface	-0.19	1.0
I-SPY2 T1-CE	no interface	-0.85	0.26
I-SPY2 T2w	no interface	+0.50	0.11

Table 3. Replacing label-style noise fill with the real-intensity remap, with the partition and region target means unchanged: change in out-of-domain Dice for every training modality of every task with a ladder. p : two-sided paired Wilcoxon on patients, pooled over that row’s held-out contrasts, Holm-corrected across all fourteen rows. (Table 10 corrects within each ladder instead, a smaller family, so a few cells significant there are not significant here.) Every large gain is on a no-tissue-interface target; no interface row gains more than 1.4 Dice. *Open-MS has eight test patients and ON-Harmony eight test sessions (four subjects, each scanned at two sites), so no row of theirs can reach $p < 0.05$ after correction even when all eight move the same way, as they do for Open-MS FLAIR (8/8 improve) and ON-Harmony T1w and T2w (8/8 worsen). For ON-Harmony we treat the session as the unit; two sessions of one subject are not fully independent, so its p -values are, if anything, optimistic. [†]31 regions, each bounded by a genuine tissue-type transition (see text). [‡]Held-out axis is cross-dataset (HaN-Seg, PDDCA), since ToothFairy2 is CBCT only. HD95 in Tab. 9.

FLAIR (+7.7), which improves for all eight test patients but cannot reach significance with eight. No interface row gains more than 1.4; CHAOS T1in and ToothFairy2 gain about one point. Pooled per task (Fig. 2), the step is worth +5.3 on Brats-GLI and +4.2 on Open-MS, against +1.0 on CHAOS, +1.4 on ToothFairy2 and -1.9 on ON-Harmony. The converse does not hold. Several appearance-defined settings show no Dice effect: Open-MS T1w, Brats-GLI T2w and FLAIR, and both Breast rows (-0.85 and +0.50, four cohorts pooled by contrast). So a large texture gain indicates an appearance-defined target, but an appearance-defined target does not guarantee one.

ON-Harmony labels 31 brain regions with the standard FreeSurfer subcortical parcellation [12]: ventricles, cortex, white matter, cerebellum, brainstem and eight deep grey nuclei. Every one is bounded by a tissue-type transition (CSF against tissue, grey against white matter), so we group it with CHAOS as an interface task. Its negative effect is

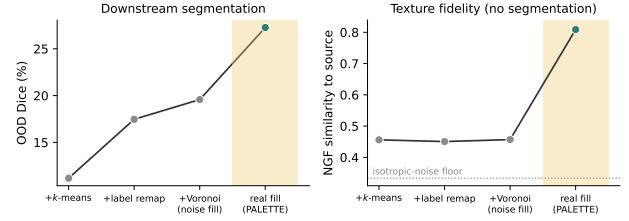


Figure 3. Noise versus real fill on Open-MS FLAIR. Left: out-of-domain Dice at each rung. Right: NGF similarity of the augmented volume to its source (mean $\pm$ s.e. over 30 scans), a segmentation-free measure of retained texture. The noise-fill rungs keep only the partition edges while Dice rises; the real-fill swap (shaded) lifts both.

Method	FLAIR	T1w
PALETTE-Aug	0.809	0.894
Auglab	0.768	0.777
SynthSeg-EM	0.462	0.422
noise fill	0.457	0.401
SynthSeg-noEM	0.330	0.319

Table 4. Texture fidelity: NGF similarity to the source scan (mean over $n=30$ Open-MS scans; $\approx 1/3$ = isotropic-noise floor). The PALETTE-Aug row is the transform alone (the real-fill rung), without the Auglab pipeline.

larger than CHAOS’s, perhaps because it labels nearly the whole brain; one densely labeled task cannot test this. Its reference labels are also made on T1w and registered to the other contrasts, which may favor some fills over others.

Does the fill swap change texture? Noise versus real fill.

The ladder shows that the fill swap changes Dice, not that it changes texture. We check this without any segmentation, with the normalized-gradient-field (NGF) similarity [16] between each augmented volume and its source scan. NGF measures how well image gradients align: it is high when edges and fine structure are kept and falls to about 1/3 for isotropic noise. Figure 3 follows NGF and Dice along the same rungs on Open-MS FLAIR. The three noise-fill rungs stay at NGF 0.45–0.46, above the noise floor because the partition edges survive but far below the source, while their Dice rises from 11.2 to 19.6. Those gains come from contrast randomization, not texture. Only the real-fill swap raises both, NGF to 0.81 and Dice to 27.3. Table 4 compares the methods on the same scale. PALETTE-Aug (the real-fill rung, without Auglab) keeps significantly more source texture than Auglab (paired Wilcoxon over the 30 scans; FLAIR $p = 3.2 \times 10^{-3}$, T1w $p = 1.2 \times 10^{-6}$), and the label-style methods stay near the floor.

Why Auglab is the closest competitor. Auglab is the strongest competitor on the cross-task test, and NGF suggests why: it keeps most of the source texture, only mostly less than PALETTE-Aug and far more than the label-style methods (Tab. 4). The large margins appear against the methods that discard texture.

What the test does and does not show. It compares two versions of one method that differ in one component, so differences in augmentation strength, partition granularity or training schedule cannot explain it. It is also separate from the benchmark ranking: on Mandible the texture step helps while the full method places third (§4.2).

Supporting analyzes. The supplement gives a component ablation of the k -means partition, the Voronoi sub-parcellation and the signed remap (Sec. 11), per-setting results for all eighteen source-modality settings, every rung of every ladder, and the HD95 version of Tab. 3 (Sec. 9).

5. Discussion and Limitations

Why the effect follows the anatomy. Liver, spleen and kidney are encapsulated organs with a clear physical boundary, and edge-following methods have long segmented them well [25, 46]. An augmentation that scrambles their internal texture but keeps the boundary loses little. Diffuse glioma has no such boundary: tumor cells infiltrate beyond both the enhancing margin and the FLAIR abnormality [17, 47], so the target must be inferred from appearance. MS lesions are likewise defined by tissue signal [56]. This account interprets the intervention of Sec. 4.3; it is not separate evidence for it.

Appearance-defined is necessary, not sufficient. Open-MS T1w, Brats-GLI T2w and FLAIR, and both Breast rows show no texture effect. We have two untested hypotheses. The effect differs sharply by training contrast within one task (Brats-GLI: +13.3 from T1c, +7.2 from T1n, about zero from T2w and FLAIR), so how much discriminative texture the training contrast carries may matter as much as the target. And Breast pools four cohorts from different trials, which may dilute a cohort-specific effect. The remaining gains on interface tasks, where texture does not help, likely come from the wider range of region-to-region contrasts, including sign inversions, that PALETTE-Aug samples.

Limitations. We assign each target to the interface or no-interface group by its anatomy, not by a measured boundary statistic. Open-MS has eight test patients and ON-Harmony eight sessions of four subjects, so their effects can be consistent without reaching significance, and ON-Harmony’s

session-level tests are optimistic. Spine was run on a separate pipeline and has no SRCSM arm. The cross-task test treats our eight tasks as fixed; with the task as the unit, the advantage over SRCSM is not significant. PALETTE-Aug targets the single-source setting and is complementary to multi-source domain generalization, which assumes several labeled source domains.

6. Conclusion

We asked when preserving real image texture matters for contrast-robust segmentation. Large gains from it occur only where the target is defined by tissue appearance rather than by an anatomical interface (+13.3 and +7.2 Dice on brain tumor, +7.7 on MS lesions, at most +1.4 on interface targets), though not on every appearance-defined target. PALETTE-Aug, which preserves texture and needs no dense label map, is the best method overall across eight tasks and sixteen datasets. The task-level account makes a testable prediction for new tasks: a large gain from texture preservation should not be expected where the target has an anatomical interface.

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PALETTE-Aug: A Simple Texture-Preserving Augmentation for Medical Imaging Domain Generalization

Supplementary Material

7. Training configuration

All methods share one nnU-Net [22] configuration; only the input augmentation differs. Nothing below is tuned per dataset.

Training schedule. 3D full-resolution nnU-Net, SGD with Nesterov momentum 0.99, initial learning rate 10^{-2} with the nnU-Net polyLR decay, 250 mini-batches per epoch, combined Dice and cross-entropy loss, deep supervision on. Patch size, spacing and batch size are whatever the nnU-Net planner selects for that dataset, identically for every method. The epoch budget is fixed per dataset and shared by all methods on it: 2500 (Brats-GLI), 2000 (ON-Harmony, Open-MS), 1000 (ToothFairy2, I-SPY2) and 200 (CHAOS, Pelvis); Spine was trained on a separate pipeline. Each method is trained as three models on three train/validation folds, all scored on the same held-out test set; folds are the same across methods.

PALETTE-Aug hyper-parameters. Set once and never tuned per dataset: k -means class count drawn per sample from $\{2, \dots, 6\}$, skipped in favor of a single global remap with probability 0.10; Voronoi sub-parcellation of each intensity class with probability 0.60 (else left as one region); signed affine coefficient $\alpha \in \pm U(0.5, 2.0)$ applied per sub-region as in Eq. (1) (always applied, to whatever partition results from the k -means/Voronoi steps); per-label affine decoupling applied independently per label with probability 0.5 (skipped for any label with under 4 foreground voxels in that sample); Gaussian blur σ drawn from $\{0, 0, 0, 0.3, 0.5, 0.8\}$, applied twice (after the region-level and after the label-level remap), so mostly not applied at either point. The transform is applied on the GPU, on-the-fly, to a fixed 50% of training iterations (*train050*); the remaining half of iterations use the un-transformed real image.

Validation-time configuration. *val000* (the configuration reported in the main paper) applies no synthetic contrast to the validation split, matching every competing method and keeping model selection like-for-like. *val100* applies PALETTE-Aug to the entire validation split; it is reported in Sec. 10.

Baseline configurations. SynthSeg-noEM uses the authors' released generator with its default priors. SynthSeg-EM is our re-implementation of the variant described but

not released in the original work; it estimates a per-class intensity model from the sparse task labels available in each dataset via expectation-maximization, and is otherwise identical in its sampling procedure. SRCSM and Auglab use their published default configurations. No competing method was re-tuned in our favor, and all use the same folds, schedule and epoch budget as above.

Compute. Training was performed on NVIDIA L40S and H100 GPUs. A single fold costs from under an hour (CHAOS, 200 epochs) to roughly three days (Brats-GLI, 2500 epochs); the full study is dominated by the six methods $\times$ eighteen settings $\times$ three folds of the main comparison, plus the fourteen ablation ladders of Sec. 4.3.

8. Datasets and evaluation protocol

All evaluation is performed on held-out test cases only. For each task the model is trained on a single source contrast and evaluated on every contrast of that task's test split, including the training contrast itself; training and validation subjects never appear in any reported number.

ON-Harmony [52] — 31-region healthy-brain segmentation. Reference labels are *silver standard*: they were produced automatically with FreeSurfer [12] on the T1w volume of each subject and then registered to that subject's other contrasts, which are acquired in the same session. They are therefore not manual annotations, and absolute Dice on this task should be read as agreement with an automated reference rather than with a human rater; the comparison between methods, which all use the identical reference, is unaffected.

Open-MS [29] — MS-lesion segmentation with manual consensus annotations, evaluated across FLAIR, T1w and T2w.

Brats2024-GLI [10] — post-treatment glioma sub-region segmentation, evaluated across T1n, T1c, T2w and T2-FLAIR.

CHAOS [25] — abdominal organ segmentation (liver, spleen, both kidneys), evaluated across the CHAOS MR contrasts and, as a cross-modality transfer, on held-out CT from AMOS [23] and SLIVER07 [19].

ToothFairy2, HaN-Seg and PDDCA [5, 38, 40] — mandible segmentation, trained on dental cone-beam CT, the one non-CT, non-MRI training modality here. ToothFairy2’s annotation coverage is cohort-dependent, so every column of this task is scored on the mandible alone (teeth excluded, matching the external references), which keeps the numbers being averaged commensurable. ToothFairy2 acquires CBCT only, so its held-out evidence is entirely cross-dataset: HaN-Seg head-and-neck CT and MR-T1 and the PDDCA CT, the two CT sources pooled as one contrast group. ToothFairy2’s volumes are stored superior-inferior inverted relative to the external cohorts — an inversion invisible to the image header, found by checking anatomy directly — so the external test volumes are flipped into ToothFairy2’s convention before inference, which leaves Dice and HD95 unchanged under the rigid transform. The correction matters for the comparison: a method without flip augmentation (SRCSM) was penalised by the inversion far more than the others, and it is after this correction that SRCSM leads this task.

Breast: I-SPY2, Duke-Breast-MRI, I-SPY1, ACRIN-6698 [13, 21, 32, 37, 41] — breast-lesion segmentation on malignant-only cohorts. I-SPY2 (CC BY 4.0) trains, on its first-post-contrast T1 (T1-CE) and its T2w; its reference is the trial’s functional tumor volume mask. Three external cohorts are test-only. Duke-Breast-MRI and I-SPY1 come through the MAMA-MIA release, whose expert primary-lesion protocol both share, and contribute T1-CE and pre-contrast columns; Duke-Breast-MRI is CC BY-NC 4.0, a non-commercial term no other cohort here carries. ACRIN-6698, the diffusion sub-study of I-SPY2 with no patient in common with our I-SPY2 subjects, contributes the baseline diffusion-weighted acquisition with the trial’s whole-tumor DWI region, a contrast neither training modality resembles. Because I-SPY2 is mostly acquired unilaterally, bilateral external volumes are cropped to the lesion side and, for the full-thorax acquisitions, also anterior-posteriorly to the breast-plus-chest-wall extent of I-SPY2’s own crops; without the second crop every method’s HD95 is dominated by the thorax. I-SPY2’s DCE series are acquired unilaterally at some sites and bilaterally at others; both field-of-view variants are kept per patient, and the significance tests treat them as one patient.

Spine [50] — vertebrae, intervertebral discs and spinal canal. Results for this task were computed on a separate pipeline with the same method suite except SRCSM, trained on CT and Dixon in-phase MRI of a CT/Dixon cohort and tested on that cohort’s held-out CT and Dixon fat, in-phase and water images and on the T1w and T2w of SPIDER. Dice and HD95 are computed on each whole structure per label (the pipeline’s `global_bin` metrics). No SRCSM

arm was run, so Spine is absent from every SRCSM comparison.

Pelvis [9, 53] — ten pelvic and hip musculoskeletal structures (both hips, sacrum, and the gluteus maximus, medius and minimus and iliopsoas on each side), from the TotalSegmentator CT (CC BY 4.0) and MRI (CC BY-NC-SA 2.0) releases. The two are unpaired cohorts, and the MRI release mixes sequences by design, so its MRI column is a sequence-heterogeneous test rather than one named contrast. CT is subsampled to 200 cases to keep training time comparable with the other tasks.

Task	Train	Test
CHAOS	16	24
ON-Harmony	84	8
Open-MS	22	8
Brats2024-GLI	630	70
ToothFairy2	408	71
I-SPY2	476	84
Pelvis (CT / MRI)	170 / 156	30 / 27

Table 5. Per-task subject counts for the training cohorts, pooled across the 3 folds used throughout (± 1 between folds). ON-Harmony counts are scan sessions; the eight test sessions are four subjects, each scanned at two sites. External cohorts are test-only and counted separately: AMOS and SLIVER07 (CHAOS), HaN-Seg and PDDCA (Mandible), Duke-Breast-MRI (291), I-SPY1 (167) and ACRIN-6698 (371) for Breast, and SPIDER (171 T1w, 219 T2w) for Spine, whose own CT/Dixon test split has 27–29 cases per contrast. I-SPY2 counts are patients; one held-out patient can yield more than one T2w image (both field-of-view variants). Pelvis MRI keeps 183 of the 298 released cases.

9. Per-setting results

Tab. 2 in the main paper reports results at the task level, pooling each task’s held-out test contrasts, all of its source modalities and its external cohorts. This section reports the same evaluation broken out into the eighteen individual source-modality settings. Every number is a cross-fold, cross-class average over all held-out contrasts of that setting, under the protocol of Sec. 4.1.

Statistical test construction. All tests are paired, and the unit of independence is the *patient*, never the fold, the image or the contrast. A patient’s score is its mean over labels and evaluated folds; where the same patient is scored in several held-out contrasts, or by the models of several training modalities, those scores move together as one unit under the null. Treating them as separate samples would be pseudo-replication and would inflate the effective sample size — a real risk here, since several tasks test a handful of patients

under many contrasts (ON-Harmony: eight sessions of four subjects, six contrasts). For ON-Harmony the unit is the session, since its four subjects leave a test that could never reach significance; two sessions of one subject are not fully independent, so its p -values are, if anything, optimistic, and we do not rest any claim on ON-Harmony alone. Per-setting comparisons (below) use the same test within each setting; the ablation ladders use the paired Wilcoxon signed-rank test [54], since there each setting is judged on its own. All tests are Holm-corrected [20] for multiple comparisons.

Why the aggregate tests are sign-flip tests on a macro-average. The aggregate tests (Tab. 2 and the per-dataset headline blocks) are not paired Wilcoxon signed-rank tests, which is the more usual choice in this literature: SynthSeg+, from the lineage we compare against, marks a method as better than its competitors using a one-sided Bonferroni-corrected Wilcoxon signed-rank test on paired Dice [4]. We therefore give our reason rather than leave it implicit. The *estimand* we test, by contrast, is not unusual at all — it is the same equal-weight-per-stratum principle the Medical Segmentation Decathlon adopted, for the same reason, as set out below.

The estimand. What we claim is $\text{macro}\Delta$: the mean, over held-out contrasts, of that contrast’s mean paired per-case difference (ours minus competitor) — equal weight per contrast, which is exactly what the *all* column of every table reports, so the test and the table measure the same thing. Equal weighting is not cosmetic here. The claim under test is cross-contrast generalization, and a contrast’s held-out case count reflects how large a public cohort happens to be, not how much that contrast should say about generalizing. CHAOS is the extreme case: its held-out CT evidence includes an external cohort contributing 200 cases, while each of its three MR contrasts contributes 8. Weighting by case count would let one CT cohort cast more than half the votes in a test whose conclusion is about transfer across MR contrasts.

This is the same problem, and the same answer, as the Medical Segmentation Decathlon’s. Facing sample sizes that “varied heavily across all tasks and target ROIs,” it ranked algorithms within each target ROI, averaged those ranks per task, and averaged the per-task ranks for the final score — choosing that over a flat average across ROIs precisely because the flat scheme means “tasks with multiple target ROIs ... would be over-weighted” [1]. Our $\text{macro}\Delta$ applies the same principle one level across, with the held-out contrast in place of the target ROI. Where we differ is only in how that weighted quantity is reached: the Decathlon arrives at it through counts of pairwise Wilcoxon outcomes at an uncorrected per-comparison α , whereas we test the weighted difference itself and Holm-correct across competitors.

Why not a pooled Wilcoxon. The signed-rank test is defined on a single undifferentiated sample of paired differences. Applying it to the concatenation of all contrasts therefore tests a case-weighted null — the imbalance above, reintroduced — and there is no standard reweighting of the signed-rank statistic that recovers the equal-per-contrast estimand. The disagreement this causes runs in both directions: a pooled test can declare a tenth of a Dice point significant because one large cohort supplies most of the cases, or miss a consistent effect confined to the smaller contrasts.

The test. Under the null that the two methods are exchangeable on a given patient, that patient’s paired differences are equally likely to carry either sign. Sign-flipping is therefore a valid randomization null for *any* statistic computed from those differences [11], $\text{macro}\Delta$ included, which is what makes it applicable where the signed-rank test is not. We report the one-sided p in the direction of the stated hypothesis, with the two-sided p retained as a secondary column, and give a 95% confidence interval from a bootstrap that resamples patients within groups of patients sharing the same set of contrasts, so each source keeps its observed size.

Evaluating the tail. The statistic is a weighted sum of per-patient contributions, so its sign-flip null distribution can be computed rather than sampled. We enumerate all 2^M sign patterns of the $M=20$ patients with the largest contributions exactly, and treat the remaining, lighter patients as a Gaussian term with their exact null variance; with twenty patients or fewer the test is therefore exact. A plain normal approximation to the whole null, which an earlier version of this analysis used, is badly conservative in exactly the designs we have, where a few patients spanning several contrasts carry most of the null variance: on the CHAOS table it returned $p = 0.013$ for a comparison that a 200,000-draw Monte-Carlo test places below 5×10^{-6} . The enumerated tail is deterministic and reaches far tails without the $1/B$ floor of resampling. Its corollary is a hard floor for small cohorts: eight patients admit $2^8 = 256$ sign patterns, so no one-sided p can fall below $1/256$ however consistent the effect.

What is reported alongside. Every generated significance report also carries a per-contrast table of mean Δ with the raw, uncorrected Wilcoxon p for each contrast individually, and the ablation ladders use the paired Wilcoxon throughout. The standard test is reported; it is simply not the one the pooled headline numbers rest on.

Contrast-group pooling. For CHAOS, the same tested modality is sometimes evidenced by more than one source: CHAOS’s CT evidence comes from its own held-out CT split plus AMOS and SLIVER07. Averaging every raw column with equal weight — the natural first implementation — silently lets a modality with more contributing sources outvote one with fewer in both the displayed *all* average

Setting	Base	noEM	EM	SRCSM	Auglab	Ours
Brats-GLI T1n	25.3*	8.7*	43.1*	40.3*	46.3	46.4
Brats-GLI T1c	28.3*	9.5*	42.9*	44.5*	46.6	47.1
Brats-GLI T2w	18.1*	8.4*	44.1*	43.8*	47.1	47.3
Brats-GLI FLAIR	21.8*	8.7*	42.3	37.5*	39.4*	43.0
CHAOS T1in	29.4*	62.6*	83.4*	83.7	83.1*	85.1
CHAOS T2spir	29.1*	56.6*	82.7*	84.1*	83.5*	84.8
ON-Harmony T1w	27.5*	66.0*	67.3*	66.4	67.6*	68.6
ON-Harmony T2w	34.8*	66.7	67.5	66.7	67.2	67.7
ON-Harmony DWI	42.6*	61.6	64.2	66.8	63.4	63.1
Open-MS FLAIR	25.2*	4.6*	32.8*	34.6*	32.8*	39.5
Open-MS T1w	16.7*	4.0*	35.1*	31.4*	35.8	37.4
Mandible CBCT	59.2*	53.6*	76.3*	79.0	78.3	77.7
Breast T1-CE	34.1*	16.6*	40.4*	18.0*	41.1*	41.9
Breast T2w	35.9*	16.5*	38.9*	14.8*	39.0*	40.5
Spine CT	27.0*	46.7*	79.1*	—	77.5*	80.2
Spine Dixon in	69.9*	48.4*	82.2*	—	84.2	84.3
Pelvis CT	35.6*	17.1*	66.5*	71.4	68.5	69.2
Pelvis MRI	24.0*	12.8*	58.6*	59.2*	60.0	62.0

Table 6. Per-setting Dice (%), best per row in bold. Each row is one training modality of one task, scored on exactly the held-out contrasts and external cohorts of its task column in Tab. 2, so the rows of a task average to that column. *: PALETTE-Aug significantly better than that method within the setting (same patient-level sign-flip test, Holm-corrected across competitors). Spine has no SRCSM run.

and the significance test, since it then supplies more of the equally-weighted columns being averaged. We instead pool disjoint sources of the *same* modality by concatenating their cases into one group before any averaging happens, then average modality groups with equal weight – for CHAOS this additionally means averaging equal-weight across liver/spleen/kidney *within* the pooled CT evidence, since AMOS’s own CT column already blends all three organs into one number before it reaches this step. The same pooling applies to Mandible (HaN-Seg and PDDCA both supply CT) and Breast (I-SPY2, Duke-Breast-MRI and I-SPY1 supply T1-CE, and Duke-Breast-MRI and I-SPY1 also supply pre-contrast); the remaining tasks have no contrast evidenced by more than one source.

Tables 6 and 7 break each task column of Tab. 2 into its training modalities. Where a margin is not significant in a single setting, the honest reading is that one setting cannot separate the methods, not that they are equivalent: several settings have few test units (eight patients on Open-MS, eight sessions on ON-Harmony), which is why the main claims rest on the task- and cross-task-level tests.

Boundary distance at task level. Tab. 8 gives the HD95 counterpart of Tab. 2. Across tasks PALETTE-Aug is significantly better than every competitor, Auglab included. Per task it has the lowest HD95 on Brats-GLI, ON-Harmony, Open-MS, Spine and Pelvis; it ties Auglab on

Setting	Base	noEM	EM	SRCSM	Auglab	Ours
Brats-GLI T1n	41.0*	78.0*	18.4*	20.8*	15.1	15.4
Brats-GLI T1c	40.3*	78.3*	19.5*	14.9	15.1	15.4
Brats-GLI T2w	54.9*	79.7*	16.5*	15.7*	14.8	15.0
Brats-GLI FLAIR	43.1*	76.0*	17.3	18.5*	19.2*	16.6
CHAOS T1in	120.0*	87.5*	33.8	26.3	43.5*	30.3
CHAOS T2spir	128.8*	92.4*	29.8	27.2	35.1*	30.9
ON-Harmony T1w	33.4*	10.4*	5.4	6.7	5.2*	4.3
ON-Harmony T2w	23.7*	4.6*	4.0	4.6*	4.2*	4.0
ON-Harmony DWI	14.7*	4.9	4.6	4.5	4.6	4.7
Open-MS FLAIR	39.9*	41.5*	23.6	23.8	26.6	24.3
Open-MS T1w	30.5*	42.3*	25.9	25.5	26.3*	24.2
Mandible CBCT	21.7	32.9*	24.9*	19.3	22.1	23.5
Breast T1-CE	61.8*	103.2*	61.7*	92.4*	57.3	59.0
Breast T2w	55.2	99.1*	60.1*	91.2*	56.9	56.3
Spine CT	78.3*	110.4*	4.5*	—	5.8*	3.9
Spine Dixon in	21.6*	113.2*	3.5*	—	3.6	3.1
Pelvis CT	146.2*	252.0*	20.0	16.0	23.5	21.6
Pelvis MRI	152.3*	271.6*	48.0	51.9*	47.8*	36.4

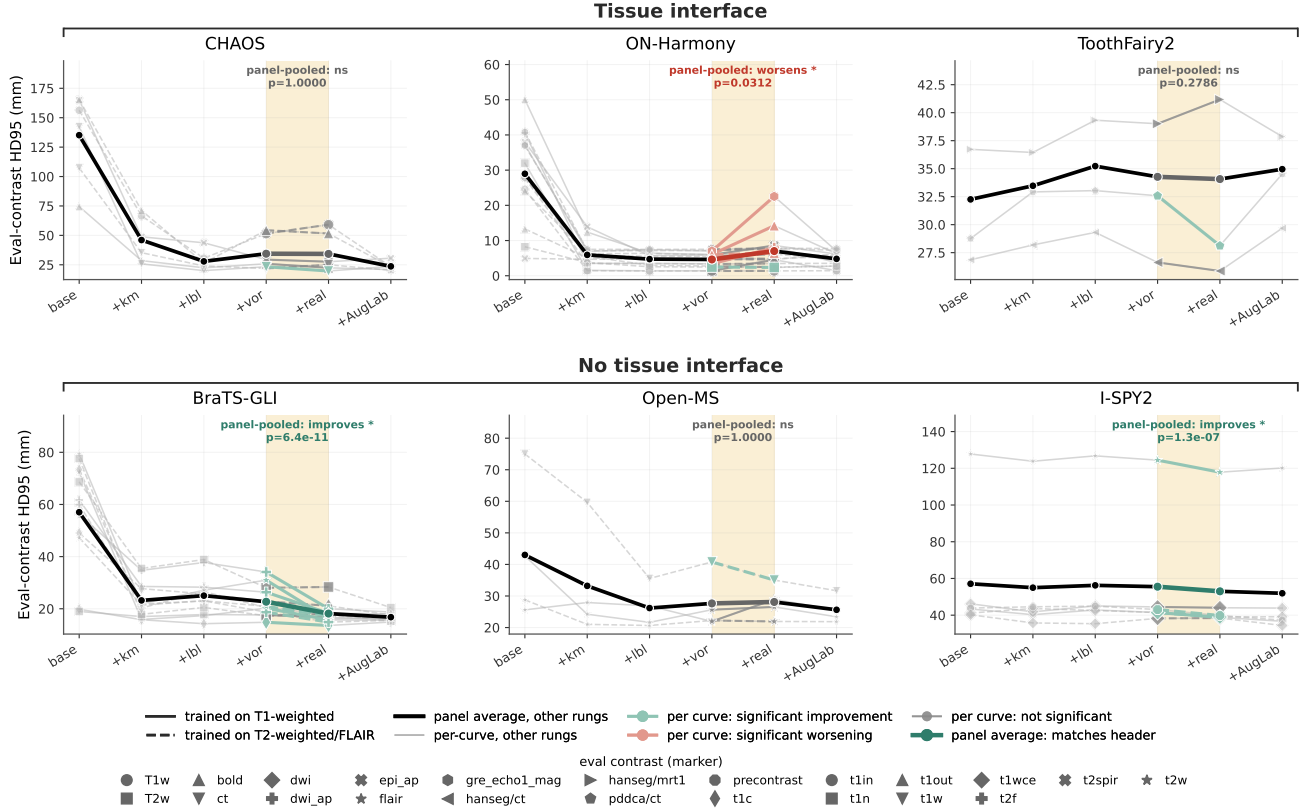
Table 7. Per-setting HD95 (mm, ↓), best per row in bold. Each row is one training modality of one task, scored on exactly the held-out contrasts and external cohorts of its task column in Tab. 2, so the rows of a task average to that column. *: PALETTE-Aug significantly better than that method within the setting (same patient-level sign-flip test, Holm-corrected across competitors). Spine has no SRCSM run.

Breast and trails SRCSM on CHAOS without either gap being significant; and on Mandible it is significantly behind SRCSM, Auglab and the no-augmentation baseline, the one task where the baseline beats PALETTE-Aug. When HD95 disagrees with Dice for a method – a noticeably better HD95 than its Dice would predict – the usual explanation is low recall: a method that predicts only a small, confident, well-localized fragment of the structure can have a deceptively good boundary distance while missing most of the target, so a strong HD95 paired with a weak Dice is a warning sign rather than evidence of competitiveness.

Full ladder: numbers and significance. Fig. 2 (main paper) shows the rung sequence visually; Tab. 10 gives the underlying out-of-domain Dice at every rung of all fourteen ladders, with significance for each step against the rung immediately before it (Holm-corrected paired Wilcoxon on patients, within each ladder’s family of steps). Most steps that add an ingredient help, as expected. Six cells are significant *decreases* (marked ↓). Two are the fill-swap step on ON-Harmony T1w and T2w, consistent with its tissue-interface grouping. One is the Voronoi sub-parcellation step on Brats-GLI T1n, which the real-fill step then more than recovers. The other three are the final +Auglab rung — on ON-Harmony DWI and twice on Mandible, the task where the complete method places third (Sec. 4.2). There PALETTE-Aug alone (the +real rung, 69.5) is the ladder’s

	Brats-GLI	CHAOS	ON-Harm.	Open-MS	Mandible	Breast	Spine	Pelvis	p vs ours
Baseline	44.8	124.4	24.0	35.2	21.7	58.5	49.9	149.3	4.8e-46
SynthSeg-noEM	78.0	89.9	6.7	41.9	32.9	101.1	111.8	261.8	1.8e-58
SynthSeg-EM	17.9	31.8	4.7	24.7	24.9	60.9	4.0	34.0	1.3e-6
SRCSM	17.5	26.7	5.3	24.6	19.3	91.8	—	33.9	3.0e-23
Auglab	16.1	39.3	4.7	26.5	22.1	57.1	4.7	35.6	8.9e-7
PALETTE-Aug (ours)	15.6	30.6	4.3	24.3	23.5	57.7	3.5	29.0	—

Table 8. Task-level HD95 (mm, ↓). Same pooling and test as Tab. 2, and likewise no cross-task average. Spine has no SRCSM run.

Figure 4. The causal-ablation ladder of Fig. 2 (main paper), same per-dataset panels, same boundary-type grouping, same fill-swap rung and same visual scheme (per-curve color only on the real-fill segment; bold black panel average; one panel-pooled p -value per panel), reported on HD95 instead of Dice (*improvement* means the fill-swap step lowers HD95). Brats-GLI and Breast improve significantly at panel level and ON-Harmony worsens; CHAOS, Open-MS and ToothFairy2 do not move significantly. Underlying per-row deltas in Tab. 9.

maximum and adding the Auglab recipe takes it back down, which places the Mandible shortfall in the wrapper rather than in the transform this paper argues for. We have not diagnosed which Auglab operator is responsible.

We likewise do not hide that Voronoi’s own marginal contribution is task-dependent rather than uniformly positive – a separate paired significance check (same protocol as Tab. 14’s all-contrast Dice, not just the OOD slice used above) confirms the same pattern: Voronoi significantly *helps* Open-MS FLAIR and (by a small margin) ON-

Harmony T1w, is flat on CHAOS T1in, and significantly *hurts* Brats-GLI T1n under the real-fill deployed configuration as well, not only here under noise fill. The full method still wins on Brats-GLI T1n regardless (Tab. 2): whatever Voronoi costs there is more than recovered by the subsequent real-intensity fill step, so the net effect of the complete pipeline is positive even though one internal component is not.

Task	boundary type	Δ HD95 (mm)	p
CHAOS T1in	interface	-2.95	0.035
CHAOS T2spir	interface	+2.64	0.88
ON-Harmony T1w	interface	+6.82	0.062
ON-Harmony T2w	interface	+0.38	0.062
ON-Harmony DWI	interface	-0.05	0.75
ToothFairy2 CBCT	interface	-0.20	0.37
Open-MS FLAIR	no interface	-3.02	0.14
Open-MS T1w	no interface	+3.94	0.88
Brats-GLI T1n	no interface	-4.59	6.8e-10
Brats-GLI T1c	no interface	-9.91	4.5e-11
Brats-GLI T2w	no interface	-1.95	0.014
Brats-GLI FLAIR	no interface	-1.71	0.21
I-SPY2 T1-CE	no interface	-3.21	9.5e-5
I-SPY2 T2w	no interface	-1.82	0.0026

Table 9. Same fill-swap step as Tab. 3 (main paper), on HD95 (negative is better), with p Holm-corrected across the fourteen rows. Five of the six significant improvements are on no-tissue-interface targets, including both Breast rows, whose Dice does not move; the sixth is CHAOS T1in. HD95 therefore leans the same way as the main paper’s one-directional Dice result, but its sign is less consistent within a task (it flips between the two modalities of Open-MS and of CHAOS), so we state the claim on Dice and report HD95 here rather than as confirmation of it.

Rung	Open-MS FLAIR	Open-MS T1w	Brats-GLI T1n	Brats-GLI T1c	Brats-GLI T2w	Brats-GLI FLAIR	CHAOS T1in
base	2.4	1.1	16.5	15.9	6.0	11.0	21.4
+km	11.2*	9.8*	38.4***	29.6***	38.6***	26.1***	84.2***
+lbl	17.5*	22.4*	41.8***	30.3	40.2**	26.4	87.7***
+vor	19.6	34.1*	39.8***↓	30.5	41.0	31.9***	87.2
+real	27.3*	34.9	47.0***	43.7***	41.8	31.7	88.2***
+AL(v000)	30.6	36.4	46.9	44.2	46.8***	40.9***	87.7
+AL(v100)	31.3	34.7	47.6*	46.5***	46.9	41.2	87.3

Rung	CHAOS T2spir	ON-Harm. T1w	ON-Harm. T2w	ON-Harm. DWI	Mandible CBCT	Breast T1-CE	Breast T2w
base	18.3	14.8	24.5	36.1	39.4	23.7	33.5
+km	79.0***	62.5*	63.5*	60.4*	64.6***	28.8***	37.0***
+lbl	83.4***	62.9	63.6	64.5*	65.6***	28.5	38.8***
+vor	82.7	63.2	64.2*	64.6	68.1***	30.8***	39.8**
+real	81.8	58.0*↓	62.6*↓	65.4*	69.5***	29.9	40.3*
+AL(v000)	86.3	64.2	63.9	60.9*↓	67.6***↓	34.7***	41.9***
+AL(v100)	87.1	64.2	64.4*	60.9	67.5***↓	—	—

Table 10. Out-of-domain Dice (%) at every rung of every ladder (Fig. 2), with significance for each step against the rung above it: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (Holm-corrected paired Wilcoxon on patients, within each ladder); ↓ marks a significant decrease. Rungs: base(line), + k -means, +label-remap, +Voronoi (noise fill), +real-intensity fill (PALETTE-Aug alone), +Auglab (val000/val100). Open-MS and ON-Harmony have eight test units each, so their smallest attainable p after correction is just under 0.05. I-SPY2 has no val100 arm. Mandible’s held-out axis is cross-dataset.

Per-label breakdown: is the fill-swap effect uniform within a task? Fig. 2 shows that Brats-GLI T2w’s pooled fill-swap effect is small and not significant, with one significantly negative eval contrast inside it (T2w→T1n). This

raises a further question: within that one (training, eval pair, does the fill-swap step affect every anatomical sub-region equally, or is the worsening itself concentrated in specific labels? We break Brats-GLI T2w→T1n down by its four labels – necrotic core (NCR), oedema (SNFH), enhancing tumor (ET), resection cavity (RC); $n = 70$ test cases for SNFH and ET, present in essentially every case (NCR $n = 51$, RC $n = 67$, since not every tumor has a necrotic core or a resection cavity).

Label	T1n→T2w	T2w→T1n	gap
NCR	21.5	19.3	+2.1
SNFH	66.3	45.3	+21.0
ET	27.5	27.9	-0.4
RC	52.3	43.1	+9.1

Table 11. Absolute OOD Dice (%) of the real-fill rung (PALETTE-Aug alone), per label, both transfer directions.

Label	T1n→T2w Δ	T2w→T1n Δ
NCR	+1.9 (p=0.10)	+0.3 (p=0.87)
SNFH	+14.7 (p=4.3e-11)	-9.8 (p=4.5e-05)
ET	+2.3 (p=0.017)	+0.9 (p=0.38)
RC	+2.9 (p=0.0003)	-6.5 (p=0.0039)

Table 12. Fill-swap step (+Voronoi noise fill → real fill), per label, both directions. p : Holm-corrected paired Wilcoxon, family = the 4 labels within each direction. **Bold** = significant.

The worsening is not a property of “the tumor” – it is concentrated almost entirely in SNFH, whose difference between directions (+21.0 Dice) is larger than the whole-tumor difference (about 14), and is the only label whose fill-swap delta significantly reverses sign between the two training directions (+14.7 when trained on T1n, -9.8 when trained on T2w). RC shows the same reversal at smaller magnitude. NCR shows no significant effect in either direction, and ET helps a little only in the direction that already works (T1n→T2w), never significantly reversing when trained on T2w. Fig. 5 shows why on one representative case: the T2w-trained model’s real-fill prediction on SNFH collapses to a thin band near the ventricle relative to its own noise-fill prediction on the same case, while its ET prediction looks essentially unchanged between the two runs.

One candidate mechanism, flagged for future work rather than tested here: a texture pattern learned from one contrast’s rendering of a diffuse, texture-defined label like oedema may not simply fail to transfer to a different contrast’s rendering of the same tissue – it may actively mislead, if the model has come to rely on a contrast-specific cue that the other contrast does not reproduce. This would be an

BraTSGLI02108101, fold 0 -- T2w-trained model evaluated on T1n

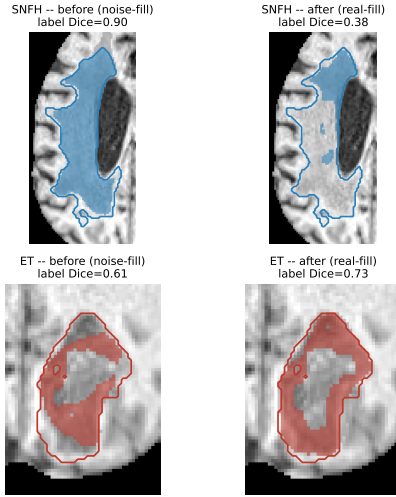


Figure 5. BraTSGLI02108101, fold 0, T2w-trained model evaluated on T1n. Zoomed crops before (noise-fill) and after (real-fill) the fill-swap step, for the SNFH-dominant (top) and ET-dominant (bottom) lesion components of the same case. Color = ground truth wherever the prediction agrees; grey = false positive; outline = ground truth.

instance of shortcut learning [14] – specifically, a network relying on a domain-specific cue that does not reproduce across acquisition protocols, the mechanism Ouyang et al. [36] identify behind failed single-source domain generalization in cross-modality and cross-sequence medical segmentation – rather than the intended, causal feature, which a simplicity-biased network need not prefer even when it would generalize better [43]. We have not tested this mechanistically here; it is the most concrete hypothesis consistent with the label-specific, direction-dependent pattern above, not a demonstrated cause.

10. Synthetic contrast at validation time

Throughout the main paper PALETTE-Aug is applied with synthetic contrast during training only (*train050/val000*), because every competing method is likewise evaluated with a real-contrast validation set; this keeps the comparison exactly like-for-like. PALETTE-Aug can additionally be applied to the validation set, so that model selection is performed under randomized contrast as well (*train050/val100*).

Validation-time randomization improves most tasks on both metrics (Tab. 13), most visibly on boundary distance for CHAOS (30.6 $\rightarrow$ 28.7 mm). The effect is not significant across tasks, and Open-MS Dice moves slightly the other way, so we do not claim it as a separate contribution; we report it because it costs nothing and because it indicates

Task	Dice $\uparrow$		HD95 (mm) $\downarrow$	
	val000	val100	val000	val100
Brats-GLI	45.9	46.6	15.6	15.1
CHAOS	85.0	85.2	30.6	28.7
ON-Harmony	66.5	66.6	4.3	4.3
Open-MS	38.5	37.9	24.3	24.2

Table 13. Applying PALETTE-Aug at validation time as well as at training time, on the four tasks where both configurations were evaluated (Mandible’s val100 arm was not evaluated on its cross-dataset test sets; the Breast, Spine and Pelvis arms were not trained). Across these tasks the two show no significant difference in either direction (same cross-task test as Tab. 2; two-sided Dice $p = 0.62$, HD95 $p = 0.18$), though val100 is significantly ahead within Brats-GLI and ON-Harmony individually. We report the like-for-like val000 configuration throughout the main paper.

that the benefit of contrast randomization extends to model selection and is not confined to the gradient updates.

11. Component ablation

To attribute the gain of the full method to its individual components, rather than to synthetic-contrast augmentation in general, we build an additive ablation ladder on top of the no-augmentation anchor (*Base*, Sec. 4.2): first the unsupervised k -means intensity partition alone (§3.1, step 1); then the anatomical label-remap step (step 4); then the Voronoi spatial sub-parcellation (step 2); and finally the full signed affine remap of Eq. (1) (step 3), which together constitute PALETTE-Aug on its own. Every rung is evaluated without Auglab, so that the ladder isolates PALETTE-Aug’s own internal components from the separate question of whether it composes with a prior augmentation pipeline, which we address next. The synthetic-data fraction is fixed at 50% of training iterations throughout, as one of several augmentation constants (alongside, e.g., the k -means class count and the affine magnitude range of §3.1) that are set once and not tuned per dataset; validation uses real-contrast data only (no synthetic contrast at validation time).

Composability with Auglab. PALETTE-Aug is deployed, throughout this paper, as one additional operator inside the Auglab pipeline rather than as a replacement for it (Sec. 4.1); Auglab is a full prior augmentation method in its own right, not a component of PALETTE-Aug, so stacking it on top of the ladder above is a composability/deployment check, not a further mechanism-ablation step. Tab. 14 keeps the two separate: the top five rows isolate PALETTE-Aug’s internal mechanism, evaluated with no Auglab throughout, while the bottom row reports PALETTE-Aug plus Auglab — the configuration used everywhere else in this paper — set apart by a rule rather than presented as the next rung of

Component	CHAOS T1in	Open-MS FLAIR	Brats-GLI T1n	ON-Harmony T1w
Base (no augmentation)	36.1	25.2	26.0	27.5
+ <i>k</i> -means partition	84.8	30.6	41.7	67.6
+ label remap	87.7	34.6	44.3	68.0
+ Voronoi sub-parcellation	87.2	36.1	42.8	68.2
+ affine remap (PALETTE-Aug alone)	87.9	40.4	48.5	63.6
+ Auglab (Ours , deployed)	87.4	39.5	47.5	68.6

Table 14. Mechanism ablation (top five rows, additive, no Auglab) and deployment composability check (bottom row). Each rung adds exactly one PALETTE-Aug component on top of the row above. Dice (%) over all test contrasts of the setting, the training contrast included, with equal weight per contrast (the all-contrast series of the same ladders as Tab. 10, which reports the held-out contrasts only).

1357 the same ladder.